



Morphometrical analysis of torrential flows-prone catchments in tropical and mountainous terrain of the Colombian Andes by machine learning techniques

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Received: 3 April 2020 / Accepted: 26 September 2020
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Abstract

Assessment of susceptibility to torrential flows at the catchment scale is a necessary step for prioritising zones that can be identified by hazard and risk maps. In the Andean region of Colombia, where these phenomena are common and highly destructive, few studies have been carried out on the subject. Considering that the effects of torrential flows will likely increase in the future due to extreme weather events and urbanisation of low prone areas, effective land management plans should include the assessment of susceptibility to this type of event. In this study, machine learning techniques were applied with morphometric parameters of mountainous catchments in the northern Colombian Andes to analyse torrential susceptibility in tropical environments. Several feature selection methods were implemented using statistics to evaluate the relationship between morphometric parameters and torrential susceptibility. The parameters that showed the strongest relationship with torrential catchments were selected for application of machine learning techniques, along with evaluation of performance and prediction ability. The analysis was carried out on morphometric parameters regarding the catchment's drainage network, basin geometry, drainage texture, and relief characteristics. According to the results, the catchments characteristics related to relief and drainage texture are the most important for the study of torrential susceptibility. Within these groups of predictor parameters, relief ratio and constant of channel maintenance are the highest ranked variables. Although the Melton index shows a good prediction capacity, thresholds proposed in the literature are not valid in the Colombian Andes.

Keywords Torrential flows · Susceptibility · Tropical · Andean range · Colombia

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1 Introduction

Torrential flows are considered highly destructive phenomena and one of the worst natural hazards of hydrometeorological origin in terms of the number of fatalities and economical loss (Alfieri et al. 2012; Doocy et al. 2013; Jonkman 2005; Marchi et al. 2010). The occurrence of torrential flows is associated with short, high-intensity rainfall in small, and mountainous catchments resulting in a short lead time for people to take actions to avoid or reduce damage to property and risk to life (Koutroulis and Tsanis 2010; Marchi et al. 2010).

According to Dittmann (1994), between 1959 and 1991, there were 3934 flood-related deaths in the USA, of which 80–90% were caused by flash floods or mudslides due to heavy rainfall, with an annual average of 119 deaths. During 1969–1981, 1185 deaths were associated with 32 events, an average of 37 deaths per flash flood or mudslide. Based on two publicly accessible global databases on natural disasters, the EM-DAT database from OFDA/CRED and the Natural Hazards Assessment Network (NATHAN) of Munich-Re, Barredo (2007) reported 47 major flood-related disasters in Europe from 1950 to 2005. Of these events, 49% are classified as flash floods that caused 2764 fatalities or 52 fatalities per year on average, which account for 40% of the flood-related casualties.

Torrential flows occurrence has increased during the last decades, and there is a high probability it will become more critical, causing an increase in human and economic loss (Balbi et al. 2013; Changnon et al. 2000; Ferri and Maber 2013; SREX 2012). This will happen as a consequence of the increase in the frequency or intensity of extreme weather events associated with global warming (Easterling et al. 2000; Huntington 2006; Morss et al. 2011), and the increase in population and property at risk from extreme weather (Changnon et al. 2000; Morss et al. 2011; O'Brien et al. 2006).

The identification of torrential flow-prone catchments is a key step in the prioritisation of areas that require detailed hazard and risk studies, or risk reduction measures. The unavailability of detailed information in remote or rural areas makes the evaluation of hazard and risk in large areas difficult. By identifying susceptible areas, stakeholders can make better decisions about prioritising resources and planning land use. Incorporating this knowledge into land use plans is crucial, considering the urban sprawl on the downstream banks in the mountainous zones. Terrain attributes obtained from digital elevation models (DEM) function as surrogates to understand complex geomorphological relationships in a simplified way (Grohmann 2004; Gritzner et al. 2001; Moore et al. 1991; Pachauri and Pant 1992). Several previous studies have used catchment morphometric variables to assess their dynamics. Earlier works that tried to integrate a large set of morphometric parameters to evaluate flash flood- and debris flow-prone catchments correspond to those proposed by Jackson et al. (1987) and Melton (1957). These works were followed by several studies in arid and high-latitude areas (Abdelkareem 2016; Bangira 2013; Bardou 2002; Bertrand et al. 2013; Chen and Yu 2011; Martins and Gadiga 2015; Tsanakas et al. 2016; Wilford et al. 2004; Xiao 1999; Youssef et al. 2011). Even though these studies in high latitudes give a good insight into the use of morphometry to evaluate debris flow hazards, it is not expected that the results can also be applied to tropical areas, because of the difference of rain patterns that determine most of the environmental and geotechnical condition of soil materials. For instance, in arid and semi-arid regions, torrential flows are reported during heavy sporadic rainfall events in winter, normally on moderate or small catchments with poor or sparse vegetation, steep slopes, poor soil development, large impermeable areas, and a fast rate of flood rise (Abdelkareem 2016; Xiao 1999). In contrast, in tropical

areas, most of these events occur during long-lasting rainfall seasons, often affecting highly vegetated areas with thick soil profiles, mostly with high permeabilities (Aristizábal et al. 2020). Nevertheless, few studies have been carried out evaluating the morphology of the basins associated with these events in tropical areas (Honer 2010; Mesa 2006; Montoya Jaramillo et al. 2009).

Although torrential flows are a common phenomenon in Colombia, no studies have presented a comprehensive analysis of torrential flow susceptibility on a catchment scale. According to the disaster database DesInventar (www.desinventar.org), 1356 small and moderate torrential flows have been registered in Colombia from 1921 to 2018, which have caused 3318 deaths. Additionally, two catastrophic torrential flows have occurred: one triggered by the eruption of the Nevado del Ruiz volcano on November 13, 1985, that resulted in more than 22,000 deaths in the towns of Armero and Chinchiná, and the other in June 6, 1994, in the Páez river catchment, triggered by a 6.4 magnitude earthquake, followed by a rainfall season that triggered 3000 landslides and a debris flow. Although torrential flows represent only 2.5% of the total disasters reported, they caused 5.6% of the fatalities and 55.9% of the fatalities of flood-related events, showing its highly hazardous and destructive nature. However, the impact data are usually underestimated because they are often reported as landslides or floods, given the lack of consistent definition or criteria for torrential flows.

In the present work, machine learning techniques have been applied with morphometric parameters of mountainous catchments in the northern Colombian Andes to analyse torrential susceptibility in tropical environments. Torrential-prone catchments were compared to similar catchments without historical torrential reports, aiming to recognise key morphological features that can be used to differentiate basins that are susceptible to torrential flows.

1.1 Torrential flows in the Colombian Andes

Differences between clear stream floods and landslides are well established physically (Bowles 1985; Coussot and Meunier 1996; O'Brien and Julien 1985; Patton 1988; Pierson and Costa 1987a, b; Takahashi 1981); however, between them there are wide and diverse types of phenomena involving a mixture of water and sediments in varying proportions that will be called torrential flows in this study.

In a broad sense, flows are defined as a mass transport phenomenon composed of a mixture of fine and coarse material with a variable quantity of water whereby both solid and fluid forces strongly influence the motion (Iverson 1997; Nettleton et al. 2005). According to the type of process dominating, flows range from gravitational hillslope processes to stream-flow water floods, with the term hyper-concentrated flow used to describe intermediate phenomena between clear stream-flow water floods and debris flows.

An important number of flow-like landslide classification have been suggested by different authors (Ancey 2001; Aulitzky 1980; Costa 1988; Coussot and Meunier 1996; Hungr et al. 2014; Iverson 1997; Jakob and Hungr 2005; Pierson and Costa 1987a, b; Takahashi 1981; Varnes 1978), where several criteria are used to differentiate types of flows composed of a mixture of water and sediment, including the number of phases, main transport agent, sediment concentration, flow velocity, material type, and the phases involved in the forces and momentum transference.

Actually, in nature exists a continuum of torrential flows between stream floods and debris flows, and boundaries between them are not sharp; torrential flows may consist of

several flow processes at different points along the flow path and at different times during the same event (Borga et al. 2014; Costa 1988; Jakob and Hungr 2005; O'Brien and Julien 1985). For this reason, Hungr (2005) proposed an approach for debris flows, very similar to the original description provided by Stiny (1910), as a concatenated phenomenon in a catchment scale, which include the initial shallow triggered landslides, with the subsequent channelised debris flows, and the final phase as a debris flood.

In this sense, in the tropical and mountainous catchments of the Colombian Andes, torrential flows as hydrometeorological concatenated processes—similar to the debris flow described by Hungr (2005)—are common. Initially, short intense rainfall of convective origin generates, in small and mountainous catchments, a rapid concentration of stream flow in the channel network as a debris flood; simultaneously, rainfall infiltration in steep hillslopes increases positive pore pressure. According to the hydrologic, geomorphometric, and geotechnical features of hillslopes, these rainstorms trigger several types of shallow or even a cluster of landslides. Then, those landslides deliver source material to channels increasing the sediment concentration and altering the fluid properties. This continuous process changes debris floods into a viscous mass surge of water and sediments that ranges from hyper-concentrated flows to debris flows as the solid concentration in flow increases. As the surge moves downstream, erosion power increases and sediments of the streambed and the conjunction of small-scale bank slides or collapses are added to the debris flow. The final deposition of these torrential flows as debris flood is often located in catchment fans that are usually highly populated resulting in huge disasters with high fatalities and economic losses. The most recent reports in Colombia include the events of Salgar on 18 May 2015 with 104 fatalities, and Mocoa on 31 March 2017, where 333 people died (García-Delgado et al. 2019).

2 Topographical and meteorological setting of the Colombian Andes mountains

Colombia is located in the northern Andes in a complex tectonic setting resulting from the Caribbean Plate moving south-eastwards relative to the South American Plate and the eastwards subduction of the Nazca Plate beneath the northern Andes along the western margin of Colombia (Kellogg et al. 1995; Taboada et al. 2000; Trenkamp et al. 2002). As a result of the oblique convergence of both the Nazca and the Caribbean Plates with the northern Andes, there is the northern Andean block, which is part of the South American Plate, moving north-westwards relative to the South American Plate (Cediel et al. 2003; Freymueller et al. 1993).

According to this tectonic setting, the Colombian Andes have been interpreted as an assemblage of terrains accreted to South America (Cediel et al. 2003). The Colombian Andes are divided into three mountain ranges. The Western Cordillera consists of an accumulation of collided oceanic blocks, mostly mafic and ultramafic island arcs accreted during the Cretaceous and early Tertiary time (Gomez et al. 2007). The Central Cordillera is characterised by the intrusion of Mesozoic rocks into a Precambrian basement and Cenozoic volcanic activity, and the Eastern Cordillera is a well-developed Palaeozoic to Cenozoic fold-and-thrust belt (Gomez et al. 2007). The Cauca-Patía graben separates the Western and Central Cordilleras, and the Magdalena Valley divides the Central and Eastern Cordilleras.

The Colombian Andean region exhibits a highly intermittent nature of rainfall in space and time, influenced by (i) topographic gradients of the three Andes mountain ranges crossing from south-west to north-east, (ii) atmospheric circulation patterns over the neighbouring tropical Pacific and Caribbean Sea, and (iii) the combined hydroclimatic and ecological dynamics of the Amazon and the Orinoco basins (Poveda et al. 2007).

The precipitation over the Colombian Andes mountains exhibits a bimodal annual cycle with a marked high rain seasons during April–May and September–November, and low-rain seasons during December–February and June–August, driven by the double displacement of the Intertropical Convergence Zone (ITCZ). The interannual rainfall variability is mainly controlled by the effects of both phases of El Niño/Southern Oscillation (ENSO): El Niño (dry phase) and La Niña (rainy phase) (Eslava 1993; Pabón et al. 2001; Poveda et al. 2007).

The diurnal cycle of maximum rainfall exhibits clear-cut diurnal (24 h) and semi-diurnal (12 h) cycles, with seasonally shifting diurnal maxima. However, these cycles are highly variable in space. Daily precipitation minima occur predominantly during the late morning hours of 09:00–11:00 LST, regardless of season and location. Over north-eastern and western Colombia, precipitation maxima occur predominantly in the afternoon. Over the western flank of the Central Andes, precipitation maxima occur either around midnight or during the afternoon, or both. And over the eastern flank of the Central Cordillera, a precipitation maximum occurs around midnight (Poveda et al. 2005). The effects of both phases of ENSO on the amplitude of the diurnal cycles are consistent throughout the Andes, as the hourly and daily precipitation decreases during El Niño and increases during La Niña (Poveda et al. 2005, 2011a, b).

These tectonically active regions with complex hydro-climatological patterns create a multi-hazard setting, where many of the most destructive disasters in the last decades have occurred (Alcantara-Ayala 2002). The Andean Mountain ranges correspond to the eastern section of the “Pacific Ring of Fire,” which extends approximately around the perimeter of the Pacific Ocean, and where much of the world’s volcanic and earthquake activity is concentrated (Anderson and Decker 1992). Moreover, the Andean Mountains influence the dynamics of weather patterns and rainfall over the region. Strong topographic features induce local atmospheric circulation and topographic peaks, which enhance deep anchored convective systems, lead to highly intense storms in space and time that trigger flash floods, debris flows, and landslides (Álvarez-Villa et al. 2011; Poveda et al. 2011a, b).

3 Data and methodology

3.1 Catchments

Morphological data concerning the major torrential flow events of 31 catchments in the period between 1950 and 2017 were collected. Descriptive information including damage, location, and date for each of the records is listed in Table 1. These catchments were selected based on disaster reports collected from the DesInventar disaster database for Colombia, which contains disaster records from 1914 to the present. Because the exact location and extra information of the disasters were often difficult to determine and/or incomplete, technical reports from governmental disaster offices and newspaper archives were also consulted online. Determining the type of event corresponding to the reports was done based on their description. Because there is a large variety of terms, often wrongly

Table 1 Selected basins of some of the most destructive torrential flow events in Colombia between 1950 and 2017

		Date	Damages	Tormentality
1	Santa Isabel Basin	23/04/1950	Multiple houses and roads affected	Torrential
2	El Hacha River	17/08/1962	122 deaths and 3000 affected people	Torrential
		4/10/1999	17 deaths, 4 missing, and more than 10,000 affected people	Debris flood
3	La Mula River	10/11/1970	36 deaths, 200 missing and 500 homeless people, 30 houses destroyed, 150 killed head of cattle, 400 m of railway, and hundreds of metres of highway lines destroyed	Torrential
4	Mulato River	29/06/1971	No damage data	Torrential
5	Rio Frio River	19/03/1974	2 deaths, 4 injured, 5 missing people, and more than 300 homeless families	Torrential
		22/02/2017	More than 2000 affected people, 600 affected and 120 destroyed houses, 2 bridges destroyed, affected aqueduct and energy supply systems	Debris flow
6	Los Andes River	31/05/1975	18 deaths, 50 houses, and 2 bridges destroyed	Torrential
7	Lejos River	9/11/1975	2 fatalities, 6 houses, and a bridge destroyed, 20 families affected	Torrential
8	El Palo River	14/11/1976	17 deaths, 30 houses destroyed, roads affected, destruction of crops, and dead domestic animals	Torrential
		11/04/2011	2 deaths, 2 houses destroyed, 7 damaged houses, 2 vehicle bridges, and commercial fishing lakes destroyed	Debris flood
9	Playonero River	25/11/1979	Unavailable official number of fatalities, ranging from 300 to 500 in unofficial reports. Damaged roads, crops, and cattle, 5000 affected people	Debris flow
		8/05/2001	3000 homeless people and aqueduct line affected	Torrential
10	Las Perlas Basin	8/04/1987	8 deaths, 24 affected people	Debris flood
11	Santo Domingo Basin	27/05/1987	12 deaths, 8 injured and 200 affected people, 10 houses and several bridges destroyed, obstruction of aqueduct. Economic losses calculated at over 500 million Colombian pesos	Torrential
12	La Ayurá Basin	14/04/1988	10 destroyed houses, 31 people affected, one vehicular and 2 pedestrian bridges and a textile factory damaged	Debris flow
13	Tabaquero Basin	13/09/1988	No damage data	Torrential
14	La Arenosa Basin	21/09/1990	4 deaths, 1200 affected, 260 evacuated, and 13 missing people, 57 affected houses, economic losses calculated at over 1000 million Colombian pesos for damages to the Calderas hydroelectric power station	Debris flow
15	La Arboleda Basin	4/06/1991	35 deaths, 2000 affected people, 60 relocated families, 20 destroyed and 40 affected houses, Destroyed local phone station, bridges, and a local school	Debris flow

Table 1 (continued)

		Date	Damages	Tormentality
16	Tapartó River	26/04/1993	120 deaths, 320 affected people, 78 affected families, 50 houses, 6 bridges, a school and 6 km of road destroyed. Economic losses calculated at 1300 million Colombian pesos	Debris flow
17	Desmontadora Basin	11/04/2015 17/12/1993	45 affected and 18 destroyed houses, 6 destroyed bridges, 215 affected people 33 deaths, 45 injured people, 50 destroyed, and 200 affected houses, more than 700 affected people, two destroyed bridges	Debris flow Debris flow
18	Frailte River	31/01/1994	19 deaths, 40 missing and 86 injured people, 428 homeless families, one destroyed bridge, wreckage of the power supply system and obstruction of the aqueduct	Debris flow
19	Llanadas Basin	13/05/1995	No damage data	Tormentality
20	La Chapa Basin	12/11/1995	2 fatalities, 2 missing people, 8 destroyed and 20 affected houses, 1 bridge destroyed	Tormentality
21	Los Llanos Basin	9/05/2002	10 destroyed and 15 affected houses, damages to the road banks, 60 evacuated people, local soccer field destroyed	Tormentality
22	San Mateo Basin	27/10/2002	6 deaths, 35 missing people, 198 families and 96 houses affected, 4 schools and several bridges destroyed	Tormentality
23	El Barro Basin	6/10/2005	42 deaths, 11 houses and local aqueduct destroyed	Debris flow
24	Las Cruces Basin	18/05/2011	12 fatalities, 80 homeless families	Tormentality
25	San Eugenio River	14/10/2011	Affected houses, roads, and local commerce	Tormentality
26	El Indio Basin	19/10/2011	50 houses destroyed	Tormentality
27	Arayanal Basin	18/07/2012	800 affected families, 600 people with no potable water supply, 54 business premises affected, damage to roads	Tormentality
28	Juan Ramos Basin	3/09/2013	3 bridges, 5 houses, 2 farms, 5 fishing lakes and a cattle ranch destroyed, economic losses estimated at 1200 million Colombian pesos	Tormentality
29	La Honda Basin	9/03/2014	Damages to crops and one pedestrian bridge destroyed	Tormentality
30	Liboriana Basin	18/05/2015	93 fatalities, 11 missing people, 67 destroyed, 33 almost destroyed and 219 affected houses, 319 families to relocate, 1 vehicle and 7 pedestrian bridges destroyed, damages to roads and aqueduct lines	Debris flow
31	Mulato River	31/03/2017	332 deaths, 77 missing people, 17 affected neighbourhoods, 22,000 homeless people, 556 business establishments, 3 aqueducts, one sewage systems, one power substation, 2 pedestrian and 5 vehicle bridges destroyed, 10 affected roads	Debris flow

The number in the first column corresponds to the location number in Fig. 1

used on the media to describe flow related phenomena, determining the nature of the event was not always possible beyond knowing it was a torrential flow. When the reports of the event included the presence of big landslides or multiple landslide events in the source area, they were considered as debris flows. In many others torrential flows, no landslides were reported and sediments from erosion, channels, and banks were added to the flow, and they were considered as debris floods.

Additionally, 42 basins with no torrential flood reports were selected. The criteria used to choose them include: (i) having similar geographic location with those basins associated with torrential flood occurrence, normally located in the same mountain range, (ii) not having any reports on any disaster database or media, (iii) having a populated area in its valley (to ensure that the absence of any event report is not caused by lack of damage or its occurrence in a remote area), and (iv) having similar topography, with similar slopes, rugosity and size to basins associated with torrential flow events. A verification procedure was carried out using Google Earth satellite images. They were used to verify historical photographs from the earliest available (normally as early as 1970) to the present, to search visible signs of torrential flood events, like fans or a cluster of landslides. Since these catchments do not present any evidence of torrential occurrence during the past, they were called no torrential catchments. Figure 1 shows the location of the basins with torrential flood reports or torrential catchments (*T*) and the no torrential catchments (*nT*) in the Colombian Andean Mountains.

3.2 Morphometric parameters

An analysis and interpretation of remote sensing images from ALOS PALSAR radiometric terrain corrected DEM with 12.5 m resolution (ASF 2014) was carried using techniques of geographic information system (GIS). The automated derivation of topographic basin data from DEM using GIS is faster, less subjective, and provides more reproducible measurements than traditional manual techniques applied to topographic maps (Abdelkareem 2016; Martz and Garbrecht 1995).

Morphometric parameters are classified into linear, areal, and relief parameters. In this study, they were grouped into drainage network, basin geometry, drainage texture, and relief characteristics and were evaluated with established mathematical equations shown in Table 2. A suite of 28 morphometric parameters for each of the 73 catchments were computed from the raster data extracted from the DEM. The different morphometric parameters were then analysed using machine learning techniques to understand which ones could be used to analyse torrential potential and contribute to hazard assessment studies in the area.

4 Morphometric features selection

In order to find the most effective combination of morphometric predictor parameters, several different methods were implemented to select and create subsets of morphometric predictor variables to apply machine learning algorithms. Feature selection deals with the analysis of the relevance of features to reduce the number of input variables and improve the performance of the model. Several strategies are applied. Filter and wrapper feature selection methods were implemented independently in this study. Wrapper methods involve the repeated application of a prediction algorithm to test several subsets of features and select the one providing maximum accuracy (Guyon et al. 2008; Iguyon and Elisseeff

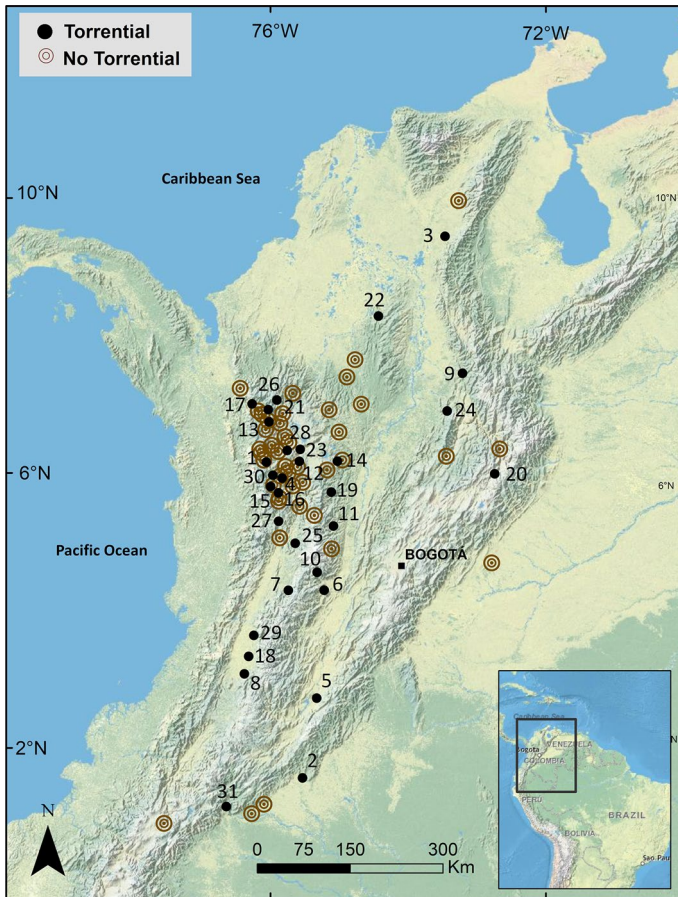


Fig. 1 Location of the studied area and the outlets of the studied catchments. Black dots correspond to torrential reports, with the number of the event corresponding to the numbers in Table 1

2003). Filter methods use statistical techniques to evaluate the relationship between each morphometric parameter and the catchment torrentiality. Those methods were complementary with the intrinsic properties of linear models and decision tree algorithms that allow obtaining a prediction capacity value for the model and information about the relevance of predictor variables (Grandvalet and Canu 2003).

Initially, a first explanatory analysis of correlation of the 29 morphometric parameters was carried out to understand their relationships (Nguyen and de la Torre 2010). Considering that morphometric parameters comprise a varying scale and the presence of outliers, all predictor variables were standardised by removing the median and scaling the data according to the first and third quartiles.

Two wrapper methods were applied: feature selector and recursive feature elimination (RFE). Feature selector is a Python class for feature detection developed by Yuan (2007). Actually, this object compiled several different feature selection methods. In this study, the feature importance was implemented using a gradient boosting machine (LightGBM) that uses a tree-based learning algorithm (Friedman 2001; Ke et al. 2017; Kraskov et al. 2004;

Table 2 Set of morphometric parameters extracted from the studied basins and their formulas

Morphometric parameter	Name	Formula	References
Drainage network			
Main stream order	S_u	Hierarchical rank	Strahler (1952)
Number of streams	N_u	$N_u = N_1 + N_2 + \dots + N_n$	Horton (1955)
Stream length	L_u (km)	$L_u = L_1 + L_2 + \dots + L_n$	Strahler (1964a, b)
Main stream length	Cl (km)	Length of longest streamline	
Stream length ratio	L_{ur}	$L_{ur} = L_u / L_{u+1}$	Strahler (1964a, b)
Bifurcation ratio	R_b	$R_b = N_u / N_{u+1}$	Strahler (1964a, b)
Ro coefficient	P	$\rho = L_{ur} / R_b$	Horton (1955)
Basin geometry			
Length of basin	L_b (Km)		Schumm (1956a, b)
Area	A (km ²)		Schumm (1956a, b)
Perimeter	P (km)		Schumm (1956a, b)
Form factor	F_f	$F_f = A / L_b^2$	Horton (1932)
Elongation ratio	R_e	$R_e = \frac{2}{L_b} * \left(\frac{A}{\pi}\right)^{0.5}$	Schumm (1956a, b)
Circularity ratio	R_c	$R_c = 4\pi A / P^2$	Miller (1953)
Compactness coefficient	C_c	$C_c = 0.2841 * P / A^{0.5}$	Gravelius (1941)
Texture ratio	R_t	$R_t = N_1 / P$	Smith (1950)
Fitness ratio	R_f	$R_f = \frac{C_l}{P}$	Melton (1957)
Wandering ratio	R_w	$R_w = \frac{C_l}{L_b}$	Smart and Surkan (1967)
Drainage texture			
Stream frequency	F_s	$F_s = N_u / A$	Horton (1955)
Drainage density	D_d	$D_d = L_u / A$	Horton (1932)
Constant of channel Maintenance	C	$C = 1 / D_d$	Schumm (1956a, b)
Drainage intensity	D_i	$D_i = F_s / D_d$	Faniran (1968)
Infiltration number	I_f	$I_f = F_s * D_d$	Faniran (1968)
Length of overland flow	L_g (km)	$L_g = A / (2 * L_u)$	Horton (1955)
Relief characteristics			
Relief	H	$H = Z - z$	Strahler (1952)
Mean basin slope	S (°)		
Relief ratio	R_h	$R_h = H / L_b$	Schumm (1956a, b)
Ruggedness number	R_n	$R_n = D_d * H$	Strahler (1964a, b)
Melton index	M	$M = \frac{H}{\sqrt{A}}$	Melton (1957)

Parameters were divided into four categories: drainage network, basin geometry, drainage texture and relief characteristics of basins

Ross 2014). The second subset was selected using the RFE method from the Feature Selection module in the Scikit-learn library. This method uses logistic regression as an external estimator. It selects features by recursively removing features and building smaller and smaller sets of features. Following this procedure, it was possible to obtain a feature selector ranking.

In a similar way, two filter methods were applied: SelectKBest and the SPSS software of IBM. The third subset of parameters was selected applying the same Feature Selection

module of the Scikit-learn library, but using the univariate statistical test SelectKBest with mutual information as the scoring function, which measures the dependency between variables. This function relies on nonparametric methods based on entropy estimation from K -nearest neighbours distances (Hosmer and Lemeshow 2000; Kraskov et al. 2004; Ross 2014). The fourth subset of parameters used IBM SPSS Statistics software. The analysis was performed using different statistical approaches such as analysis of variance (ANOVA) as a preliminary test to detect if the groups T and nT differ in the selected classification variables. The analysis allows testing of the hypothesis of equality of means between the groups in each variable. This analysis includes the univariate Wilk's Lambda statistic and p value. The method of selecting variables for the discriminant analysis was the stepwise variable selection procedure (Hosmer and Lemeshow 2000), where variables are entered and removed one at a time from the discriminant function until the most significant model has been generated based on its F value. The stepwise method selects variables by minimising the sum of unexplained variation (residual variance) between groups previously identified by grouping variables T and nT .

5 Machine learning techniques

Machine learning techniques have been used extensively during the last years for complex problems in a variety of fields, where a wide number and types of predictor variables have been explored. Machine learning is a collection of algorithms and techniques used to create computational systems that learn from data in order to make predictions and inferences (Bishop 2006; Hastie et al. 2009). The algorithms learn or extract knowledge from data in the context of a hypothetical target function (f) that best maps input or independent variables (X_n), to predict an output or dependent variable (y): $y = f(x)$.

For this kind of analysis, where the output variable (y) is known and corresponds to a categorical variable, T and nT catchments, it is named a dichotomic supervised classification problem. However, unsupervised machine learning algorithms were applied as well to describe the associations and patterns among the input morphometric variables.

There are many machine learning algorithms available today. Considering the interpretability of the model and prediction capacity, we applied a total of eight different machine learning algorithms that represent different strategies for learning the mapping function. The algorithms used were classified into three types: (i) linear unsupervised parametric algorithms: principal component analysis (PCA) (Pearson 1901) and clustering (C) (Everitt 2011), (ii) linear supervised parametric algorithms: linear discriminant analysis (LDA) (McLachlan 2004) and logistic regression (LR) (Hosmer and Lemeshow 2000; Moguerza and Muñoz 2006), and (iii) nonlinear supervised nonparametric algorithms: K -nearest neighbour (KNN) (Cover and Hart 1967), support vector machine (SVM) (Moguerza and Muñoz 2006; Vapnik 1998), and ensemble algorithms, specifically a bagging method: random forest (RF) (Breiman 2001; Gordon et al. 1984), and a boosting method: stochastic gradient boosting (SGB) (Friedman 1999).

Parametric algorithms simplify the learning process from the training data using a known function, which corresponds to a linear combination of the input variables, obtained as a result of the coefficients for the function. When the underlying function is not a linear function, nonparametric algorithms are recommended because they do not make strong assumptions about the form of the mapping target function, learning any function shape from the training data (Bishop 2006; Flach 2012; Hastie et al. 2009).

A subset of the best morphometric parameters was selected for each method, and the eight machine learning algorithms were applied to select the best model: the one composed of the best performance algorithm and the best parameter combination. The best model is the one that maximises the accuracy of the classification score. For the implementation of the machine learning algorithms for each set of morphometric parameters, the dataset was randomly split into two sets: one for training (80%) and the other for testing (20%). The training dataset was used to find the parameters that give the best performance of the models. To avoid overfitting during this step, the Scikit-learn library provides a random permutation cross validator tool called ShuffleSplit with five folds. This tool randomly samples the training dataset during each iteration to generate a training set and a validation set. This procedure on the training dataset permits selection of the best morphometric parameters. The test dataset was used to evaluate the performance and prediction capacity of the model selected with the best morphometric parameters. All the algorithms were parameterised using the method GridSearchCV of Scikit-learn library, except for LDA, which has no hyperparameters. For each of the algorithms, the following parameters were considered by optimizing their values: for LR, the inverse of regularisation strength (C) and the solver algorithm used in the optimisation problem. For KNN, the leaf passed to BallTree or KDTree, the number of neighbours to use as defaults for neighbour queries, and the power parameter for the Minkowski metric. For SVM, the Kernel coefficient (gamma), the regularisation parameter (C), and the Kernel type. For RF, the number of trees in the forest, the number of features to consider, the maximum depth of the tree, and the function to measure the quality of a split (Gini or entropy). And finally, for SGB, the learning rate, the minimum number of samples required to split an internal node, the minimum samples at a leaf node, the maximum depth of individual regression estimators, the number of features to consider when looking for the best split, the function to measure the quality of a split, the fraction of samples to be used for fitting the individual base learners, and the number of boosting stages to perform.

6 Results

6.1 Exploratory analysis of morphometric parameters

Figure 2 shows the kernel density estimation (KDE) with the probability density function for all the morphometric parameters. Parameters like stream length (L_u), stream length ratio (L_{ur}), bifurcation ratio (R_b), Ro coefficient (ρ), length of basin (L_b), area (A) and perimeter (P), show a symmetric unimodal distribution with a small spread and extreme values to the right. Parameters like form factor (F_f), elongation ratio (R_e), circularity ratio (R_c), compactness coefficient (C_c), relief (H), stream frequency (F_s) and Melton index (M), show a more spread unimodal distribution, slightly skewed with extreme values to the right. Wandering ratio (R_w), slope (S) and fitness ratio (R_f) show a unimodal distribution slightly skewed with extreme values to the left. And texture ratio (R_t), drainage density (D_d), constant of channel maintenance (C), drainage intensity (D_i) and length of overland flow (L_g) show a bimodal distribution, whilst relief ratio (R_h), ruggedness number (R_n) and Melton index (M) show a unimodal spread and positive skewed distribution.

Figure 3 shows the boxplots of the drainage network parameters grouped by T and nT catchments. Drainage network properties measure the plane arrangement of streams on the land surface by a drainage system. These aggregates are determined

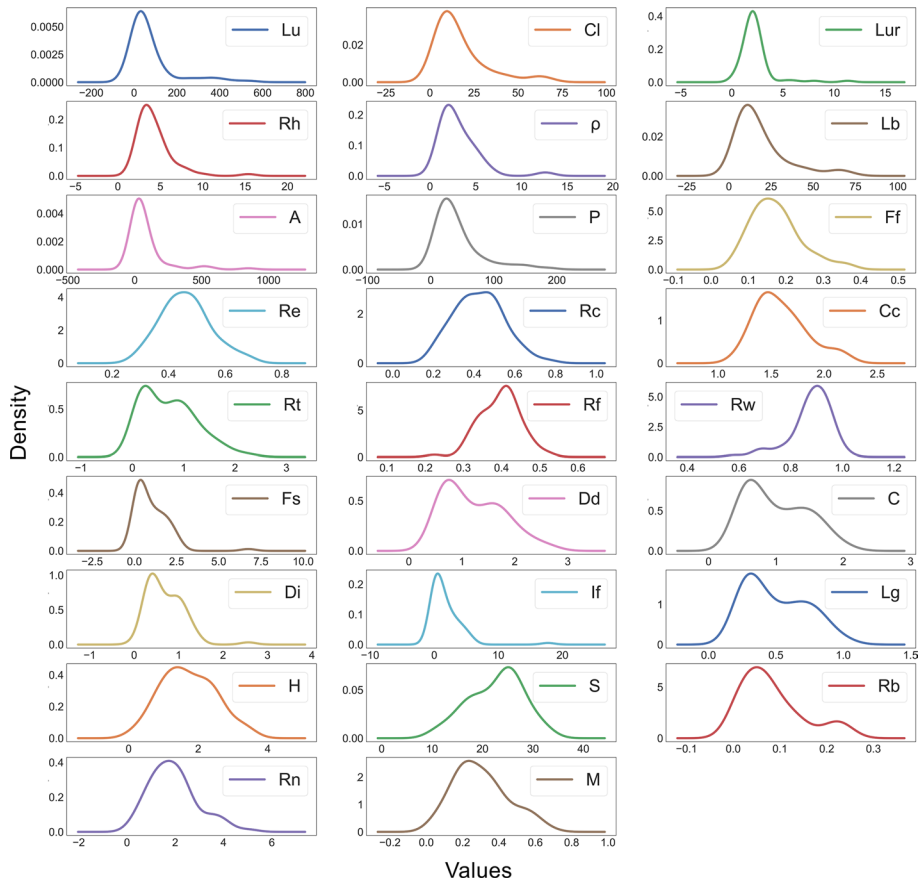


Fig. 2 Kernel density estimation (KDE) with the probability density function for all the morphometric parameters

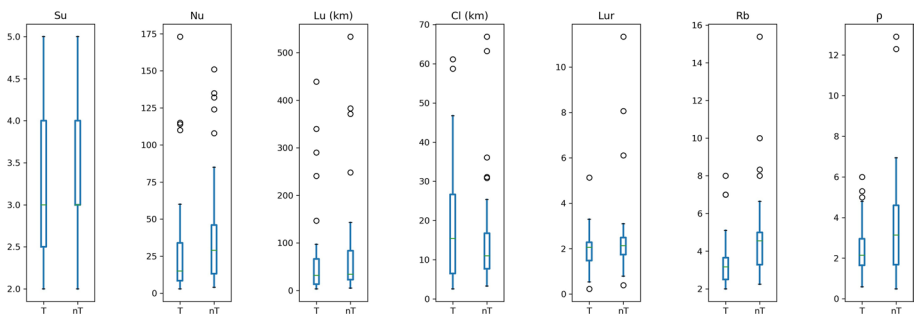


Fig. 3 Boxplots for drainage network parameters grouped by *T* and *nT* catchments

by physiographic and geologic conditions such as rock type, tectonics, and structures (Rao 2018). The main stream order (S_u) of the analysed catchments ranges from second to fifth order, with the number of streams (N_u) up to 173. As several authors have

identified, the main stream order and number of streams of the basins show an inverse linear relationship (Horton 1955). The stream length (L_u) varies from 3.50 to 533.6 km, with a maximum length of main stream (CI) of 66.90 km. Dimensionless parameters, not related to the size but to the ratio of drainages of different order, include stream length ratio (L_{ur}), bifurcation ratio (R_b), and Ro coefficient (ρ). Those morphometric parameters show, in general, lower values for T catchments (L_{ur} : 1.98, R_b : 3.5, ρ : 2.56).

Basin geometry parameters are linear properties with length and aerial dimensions required to measure the planar geometry of the basin from a plan view and their relationship with linear drainage features (Rao 2018; Youssef et al. 2011; Zavoianu 1985). Basin geometry parameters include length of basin (L_b), area (A), and perimeter (P) of the catchment, all measured in its horizontal projections in kilometres. All those parameters have similar values for T and nT, with slightly higher area and perimeter values for T catchments. Figure 4 shows these variables boxplots grouped by T and nT catchments.

All these parameters, although highly important for analysing basin morphometry, may lead to loss of accuracy due to differences in scale, shapes, and sizes of studied watersheds. Dimensionless parameters that are independent of the size and scales were computed to analyse the basin shapes. Form factor (F_f), elongation ratio (R_e), circularity

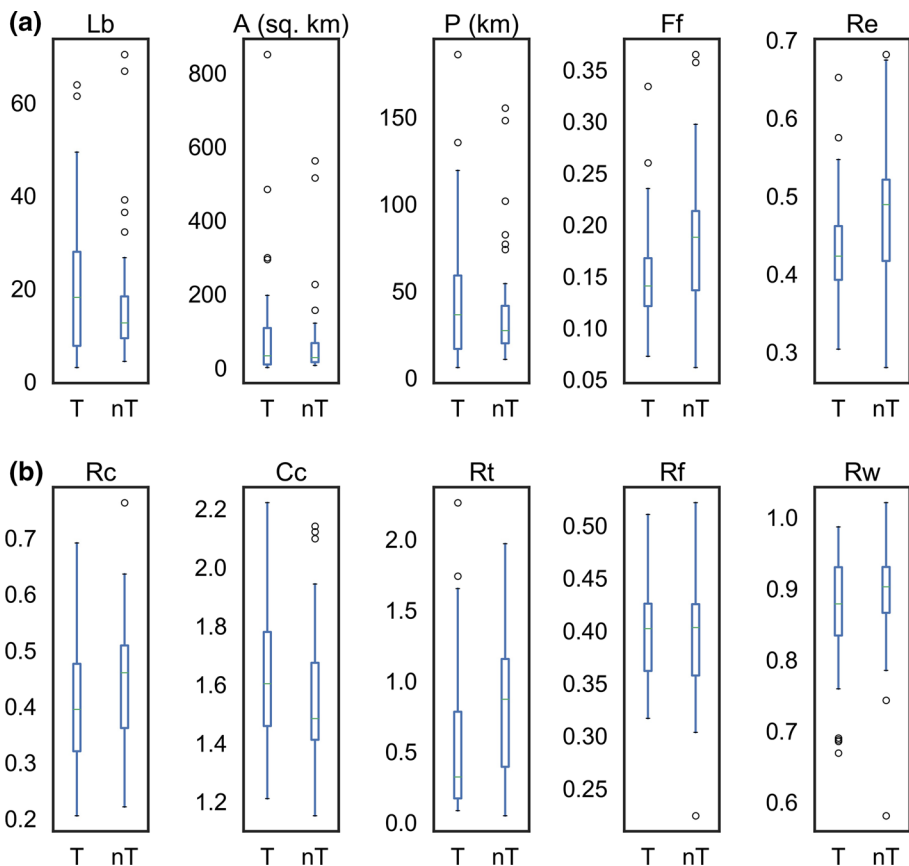


Fig. 4 Boxplots for basin geometry parameters grouped by T and nT catchments

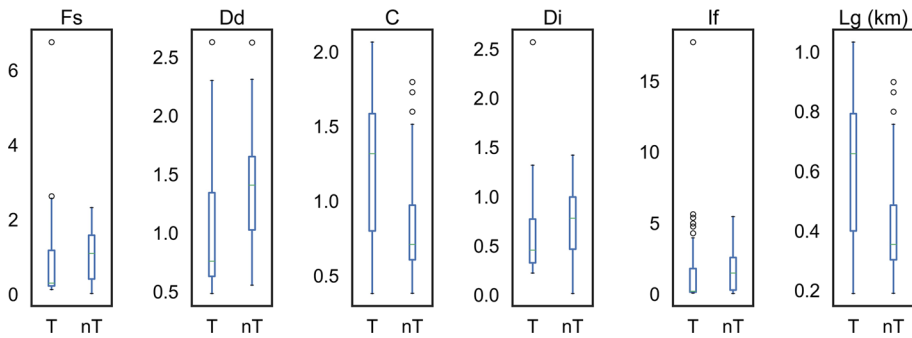


Fig. 5 Boxplots for drainage texture parameters grouped by *T* and *nT* catchments

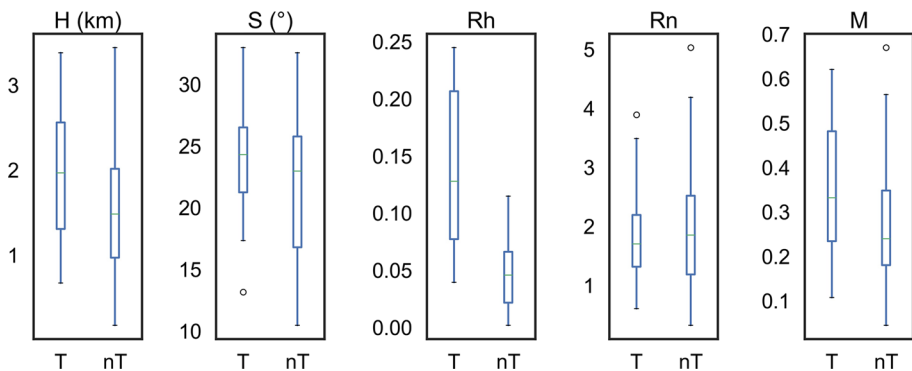


Fig. 6 Boxplots for relief characteristics parameters grouped by *T* and *nT* catchments

ratio (R_c), and compactness coefficient (C_c) show very similar ranges for *T* and *nT* catchments.

Other basin geometry parameters include not only the analysis of the basin shape and size, but also its relationship to the drainage network arrangement. Texture ratio (R_t), computed here as the number of first-order streams, shows a shorter range for *nT* catchments (0.49–1.9). Other parameters of this type, like fitness ratio (R_f) and wandering ratio (R_w), show similar values for both types of catchments.

Drainage texture parameters analyse the relationship between aerial and drainage network features, indicating the relative spacing of streams across the terrain (Dingman 2009; Rai et al. 2017; Strahler 1964a, b; Sukristiyanti et al. 2018). In general, *T* catchments show lower mean values for stream frequency (F_s) (0.94 streams per km²), drainage density (D_d) (1.06 km/km²), and drainage intensity (D_i) (0.63), and higher mean values of constant of channel maintenance (C) (1.21), Infiltration Number (I_f) (1.76), and Length of overland flow (L_g) (0.6) as shown in the boxplots in Fig. 5.

Relief characteristics include the analysis of the vertical distances traversed by drainage basins (Zavoianu 1985). Relief characteristics boxplots are shown in Fig. 6. In general, the relief characteristics of *T* catchments show higher mean values. The differences in relief (H) and mean slope (S) are less notorious, but they may be distorted by the varying scale of analysis. The relief ratio (R_h) allows a fair comparison of the relative relief regardless of

scale differences, and the differences between groups of basins are more notorious, with higher values for T . The ruggedness Number (R_n) is the relief parameter with less clear difference between groups. Finally, the Melton index (M) values for T have a broader range and higher mean values.

6.2 Multicollinearity analysis

Figure 7 shows the correlations between morphometric predictor variables using the Spearman's rank coefficient. When dealing with machine learning analysis, the performance of the model is likely to be poor when it is carried out using variables with high correlations (Dillon and Goldstein 1984). Multicollinearity does not permit to ascertain how important a feature is to the target variable.

Parameters showing high positive correlation include some basin geometry metrics like area, length of basin, and perimeter, which correlate highly with drainage network parameters like length of the basin and length of main stream. Some basin geometry parameters correlate strongly with each other like area, perimeter, basin length and elongation, and circularity ratio. Other drainage texture parameters also correlate with each other like drainage density, drainage intensity, and infiltration number. Parameters with a strong negative

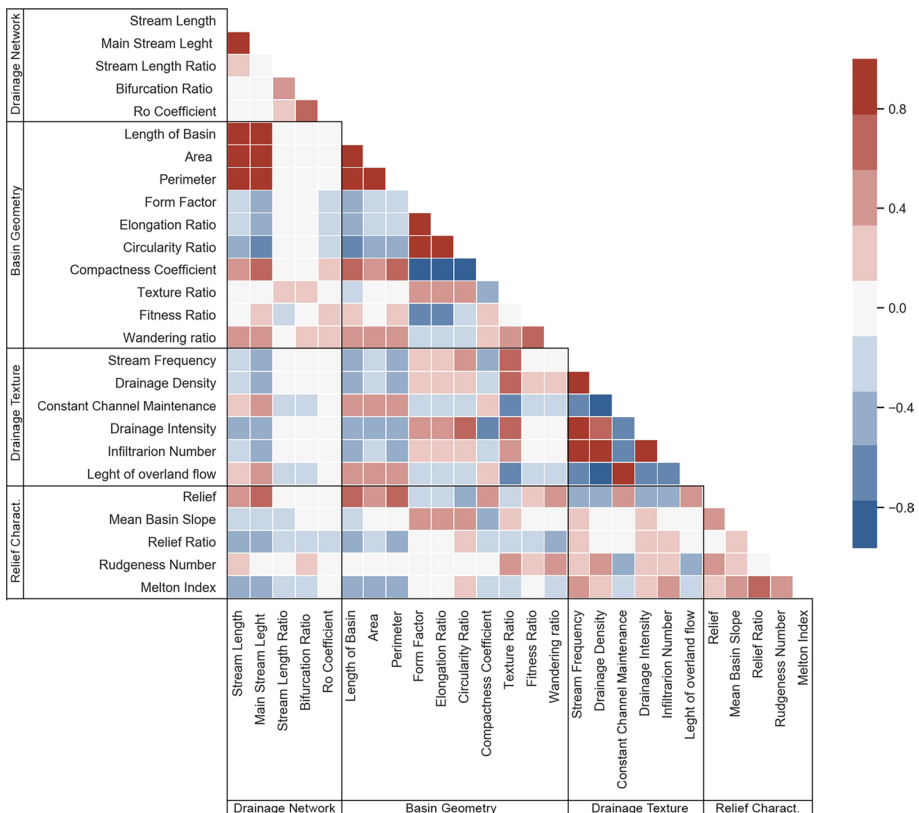


Fig. 7 Correlation matrix of the morphometric parameter

correlation include drainage density with constant of channel maintenance and length of overland flow.

Figure 8 plots each morphometric parameter uniformly around the circumference of a circle, with T and nT records located within the circle as a function of their relative attraction to each parameter. This plot is useful to detect separability between classes. The higher the value for a particular morphometric parameter, the closer the projected point will be to the corresponding anchor. There is a clear presence of two clusters, T catchments towards higher values of relief characteristics and nT catchments towards higher values of basin geometry.

6.3 Morphometric parameters selection

Table 3 shows the result of the feature selection procedure using the three tools provided by the Scikit-learn library in Python: FeatureSelector, RFE, and SelectKbest, respectively, and the discriminatory selection of the parameters using the statistical package SPSS. In the FeatureSelector method, relief ratio was selected as the most important predictor variable with a very high value, followed by form factor and stream length. In the RFE, the best parameters are relief ratio, relief, and Melton index. However, Melton index, texture ratio and constant of channel maintenance were not considered in this subset for the machine learning analysis given their collinearity with other highly important parameters like relief ratio, form factor, and length of overland flow, respectively.

In the SelectKbest method, the best parameters are relief ratio, infiltration number and stream frequency. Length of overland flow, stream frequency, drainage density and Melton

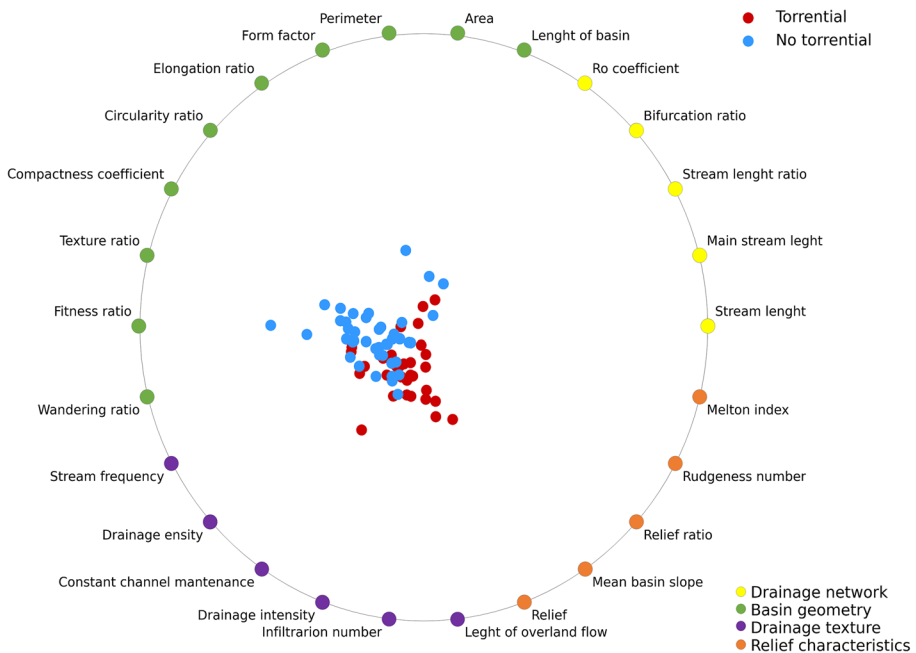


Fig. 8 Circular correlation plot for all variables with T and nT catchments

Table 3 FeatureSelector, RFE, SelectKbest and discriminant analysis results for all parameters

Morphometric parameters	SciKit-learn library			SPSS		
	Select Feature (Set 1)	RFE (Set 2)	SelectK Best (Set 3)	Wilk's Lambda	Standardised discriminant weight	Significance (Set 4)
Drainage network						
Stream length	17.15	17	1.23	1	− 0.05	0.88
Main stream length	2.65	20	3.14			
Stream length ratio	6.5	25	0	0.97	− 0.11	0.17
Bifurcation ratio	4.25	8	0	0.9	0	0
Ro coefficient	13.9	13	0	0.95	0.09	0.07
Basin geometry						
Length of basin	1.55	18	3.05			
Area	7.05	24	11.16			
Perimeter	8.4	16	0			
Form factor	26.65	4	5.19			
Elongation ratio	0	12	5.27	0.93	− 0.17	0.03
Circularity ratio	0.5	26	0			
Compactness coefficient	0.05	21	0			
Texture ratio	2.3	6	2.87	0.93	− 0.15	0.02
Fitness ratio	7.95	15	0	0.99	0.05	0.63
Wandering ratio	4.05	19	0.87	0.97	− 0.06	0.16
Drainage texture						
Stream frequency	1.9	22	16.9			
Drainage density	5.3	23	10.88			
Constant of channel maintenance	10.6	7	10.49	0.84	0.16	0
Drainage intensity	9.55	11	3.5			
Infiltration number	7.7	10	22	1	0.04	0.9
Length of overland flow	0	5	10.5			
Relief characteristics						
Relief	4.8	2	9.6	0.9	0.13	0
Mean basin slope	13.5	9	0	0.96	0.13	0.08
Relief ratio	79.75	1	31.78			
Ruggedness number	6.75	14	0	0.99	− 0.08	0.44
Melton index	4.3	3	9.45	0.28	0.63	0

 Selected variables
 Discarded variables

Light gray shaded cells indicate variables selected by the method to create each subset, and dark gray shaded cells indicate variables that were rejected based on the collinearity analysis

index were not included into the analysis because of their collinearity with constant of channel maintenance, infiltration number and relief ratio, also important parameters.

In the discriminatory selection, the Wilk's Lambda test was used for the selection of the best discriminant variables. Wilk's Lambda is defined as the percent variance in dependent variables not explained by differences in levels of the independent variable. A small Wilk's Lambda values indicate that the variable is better at discriminating between groups. The p value (significance) was calculated for the hypothesis of the equality of the means between groups for each variable. If the p value is >0.05 , the hypothesis gets rejected, meaning the variance of the groups are the same, and therefore, the variable does not have a discriminatory power. The F value for a variable indicates its statistical significance in the discrimination between groups, and it is a measure of the extent to which a variable makes a unique contribution to the prediction of group membership. A variable is entered into the model if its F value is greater than the entry value and is removed if the F value is less than the Removal value. In this case, the analysis was carried out two times: first, using all the morphometric variables. When two variables have a correlation coefficient less than -0.7 or greater than 0.7 , their Wilk's Lambda value was compared, and the variable with the lowest significance was discarded. The analysis is then carried out again with the remaining variables that are independent from each other.

The discriminatory analysis reveals differences between T and nT catchments on the following morphometric parameters: Melton index, constant of channel maintenance, wandering ratio, and relief. The magnitudes of the standardised discriminant weights are shown in Table 3. The coefficients indicate the contribution of each variable to a discriminant function. Melton index is the most influential variable with a weight of 0.637 , followed by the constant of channel maintenance with 0.169 and relief with 0.132 . Positive discriminant coefficients are associated with T catchments and negative ones with nT catchments.

The ranking of variable importance that was obtained is irregular according to prediction methods, yet the highest ranked morphometric parameters were generally consistent. Relief ratio and constant of channel maintenance were the most common variables ranking high in terms of relative importance in all methods. Similarly, Melton index and length of overland flow show a high prediction capacity; however, they show a moderate and strong positive correlation with the latter. Additional to those two morphometric parameters, relief and form factor show moderate importance capacity in several selection methods, and they do not show correlation to any other metric. Figure 9 shows the histogram for each of these parameters grouped by T and nT catchments.

The relief ratio (R_h) is the ratio between the relief of a catchment and its longest dimension, measured parallel to the main drainage line. According to Schumm (1956a; b), R_h is closely related to sediment loss per unit area, where high values of R_h usually correspond to less basement rocks in the catchment and a high degree of slope. nT is highly constrained to lower R_h values, whilst T catchments are more spread into higher values, suggesting that larger height differences in shorter basins increases their susceptibility to torrential episodes. (Fig. 9a).

Constant of channel maintenance (C) is defined as the ratio between area and stream length, representing the area in squared kilometres required to maintain one kilometre of drainage channel (Schumm 1956a, b). nT is also constrained to smaller values of C , whilst T catchments have more variability, with higher values. This suggests that the development of larger drainage networks decreases its susceptibility to torrential events (Fig. 9b).

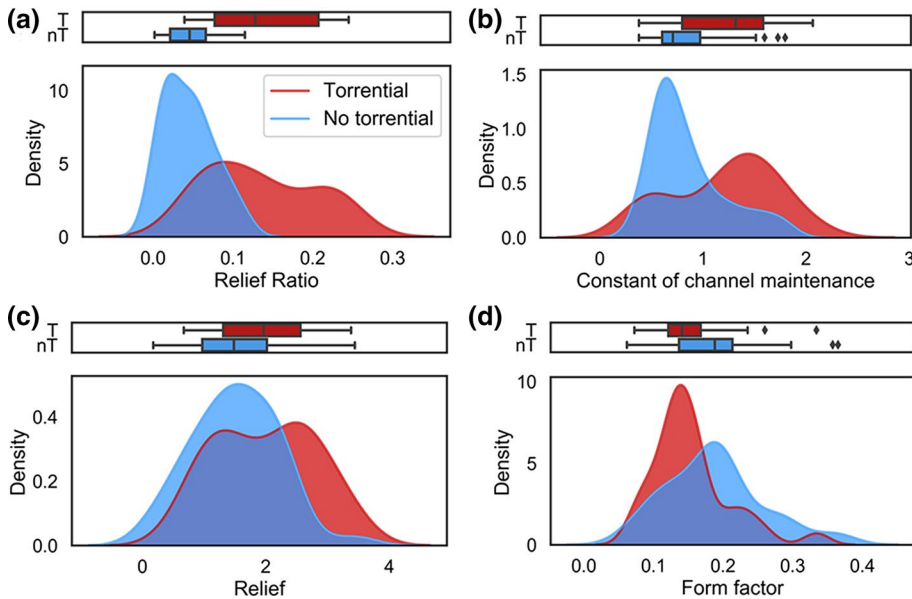


Fig. 9 Distribution of independent morphometric variables with influence on the characterisation of basin character. **a** Relief ratio; **b** constant of channel maintenance; **c** relief; **d** form factor

Relief (H) is the difference between the minimum and the maximum height above sea level of the basin in kilometres (Strahler 1952). As expected, T has higher mean H values than nT catchments, although their ranges are mostly overlapping (Fig. 9c). Form factor (Ff) is defined as the ratio of the catchment area to the square of the catchment length (Horton 1932). The smaller the value of Ff , the more elongated the catchment. The mean value of Ff for T is lower than for nT , suggesting that T catchments tend to be slightly more elongated than nT (Fig. 9d).

Using the best subset of parameters, a principal component analysis was carried out. Principal components transform the dataset into a new, lower-dimensional subspace, where the components are orthogonal and correspond to a new coordinate system created by a linear combination of the original morphometric parameters (Pearson 1901). Figure 10 shows the nT and T catchments projected onto the first and second principal components, where a clear division of the two groups can be made. Also, the original variables are plotted, showing their independence with each other (except for constant of channel maintenance and relief that showed similar information on Principal Component 1). PC1 and PC2 explained 73% of total variance in the model. For PC1, form factor has the highest positive proportion of variance (0.87), whilst relief ratio has the highest positive proportion of variance explained on PC2.

The most consistently lower-ranked variables were circularity ratio, compactness coefficient and stream length ratio. Considering the morphometric parameter groups, drainage Network parameters showed the worst performance.

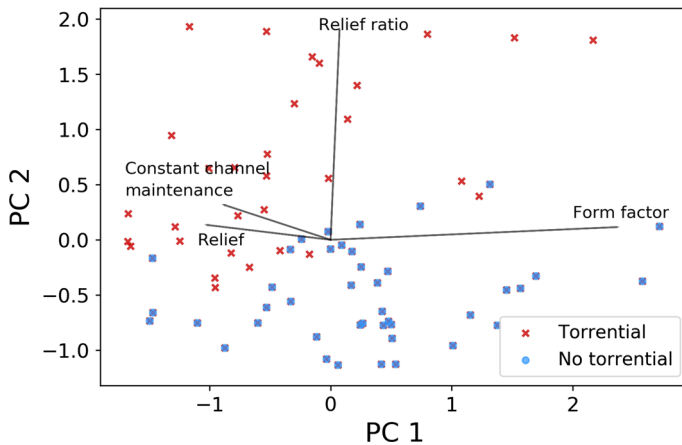


Fig. 10 Graphic showing principal component analysis using the best predictor variables

6.4 Machine learning results

Table 4 shows the accuracy values obtained for each machine learning algorithm and subset of morphometric parameters using the training data. The best morphometric predictors (relief ratio, constant of channel maintenance, relief, and form factor) were selected to build a new fifth subset for machine learning analysis. The best subset of parameters with the highest accuracy for training data are Subset 2 and Subset 5. Both models have similar morphometric parameters, considering that length of overland flow has a correlation of 1.0 with constant of channel maintenance. Only bifurcation Ratio is not included in Subset 5. However, considering the performance for both subsets in the machine learning models, bifurcation ratio only improved the performance of SGB, and for the rest of the models, the performance decreased.

Linear and nonlinear models can be used to evaluate the relative importance of morphometric predictors. Logistic regression and discriminant analysis permit interpretation of the size of the coefficients as they relate to the independent variables. For nonlinear methods, a coefficient was not obtained; however, decision trees-based models calculate the feature importance score by the average decrease in accuracy or number of times the feature

Table 4 Performance for all subsets

	SET 1		SET 2		SET 3		SET 4		SET 5	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std
LAD	0.9	0.1	1.0	0.7	0.9	0.1	0.8	0.2	1.0	0.1
RL	0.9	0.1	0.9	0.1	0.9	0.2	0.8	0.2	1.0	0.7
KNN	0.8	0.1	0.8	0.2	0.9	0.1	0.9	0.1	0.9	0.1
SVC	0.9	0.1	0.9	0.2	0.9	0.1	0.9	0.1	0.9	0.1
RF	0.9	0.1	0.9	0.1	0.8	0.2	0.8	0.1	0.9	0.1
SGD	0.8	0.1	1.0	0.1	0.8	0.3	0.8	0.1	0.9	0.2
Mean	0.9	0.1	0.9	0.2	0.9	0.1	0.8	0.1	0.9	0.2

is used in the model (Kazemitabar et al. 2017). For the linear models, relief ratio always obtained the highest coefficient value, followed by constant of channel maintenance. For nonlinear models, importance values varied considerably; however, relief ratio and constant of channel maintenance always obtained the highest values. Only in Subset 1 with SGD did drainage density show the highest importance value. The other morphometric variables selected in the methods, such as mean slope, wandering ratio, number of streams, and infiltration number, showed very low values of importance.

Using Subset 5, the best performance model corresponds to linear models, LDA (0.97) and RL (0.97), respectively, but LDA shows a lower standard deviation (0.07), indicating a better performance. Nonlinear models show a good performance (0.87–0.9), but slightly lower than linear models. The best models (LDA and LR) with the subset of best morphometric predictors show a similar prediction capacity using the test data (0.87). This means that the prediction capacity decreased, but kept a good value.

Table 5 shows the classification report and the confusion matrix that evaluate the prediction capacity of the LDA model with 15 catchments not used for the training data. Precision is the ratio between the number of true positives and all the positives given by the model. Precision for T is higher than for nT catchments, which means that the model can identify all the nT but fails to classify some of the T catchments. Considering this condition, the $f1$ -score, which corresponds to the weighted harmonic mean of the precision and recall, must be considered as the best statistic to evaluate the prediction capacity.

Figure 11 shows the learning curve for LDA with Subset 5 indicating the very good learning capacity of the model. Learning curves are used to diagnose problems with learning and the possible generalisation behaviour of the model with new records, showing how errors change as the training and validation set size increase. Whilst the validation curve is increasing slowly during the learning processes of the model with more records, the training curve slowly decreases, and the score shows a minimal gap between the two final score curves. This gap is referred as the generalisation gap. Although for the training and testing data the model shows high rate scores, the slightly increasing tendency of the validation curve indicates that the model is capable of further learning and further improvements with more record data. The high values for training and testing samples indicate a low bias, and the small generalisation gap indicates a low variance of the model's performance.

7 Discussion

The results reveal that morphometric parameters related to relief characteristics and drainage texture have the highest importance for distinguishing T and nT catchments, indicating a dominance in the torrential flow susceptibility over drainage network and basin geometry properties. For the latter groups, most of the morphometric parameters showed a low importance ranking, and only bifurcation ratio and form factor showed a moderate prediction capacity.

Table 5 Classification report and confusion matrix of the model

	Precision	Recall	$F1$ -score	Support
nT	0.80	1.00	0.89	8.00
T	1.00	0.71	0.83	7.00
Average/total	0.89	0.87	0.86	15.00

Last row shows average for first three columns and total for last

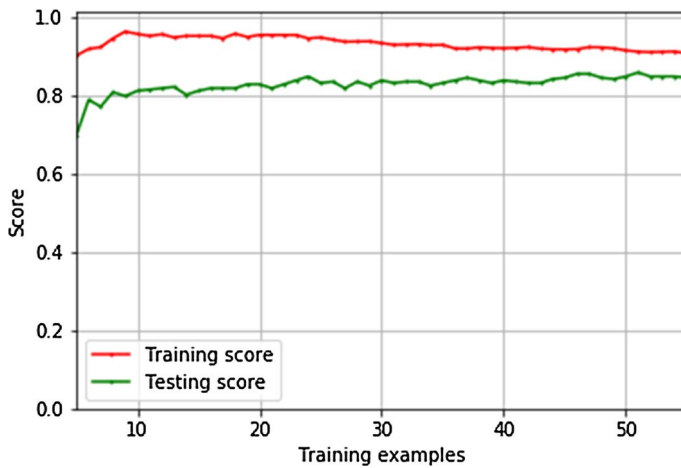


Fig. 11 LDA learning curve for Subset 5

In terms of drainage network, *T* catchments showed relatively lower mean values for number of streams, stream length, and stream length ratio, but a higher mean value for main stream length, indicating a young stage of geomorphologic development of streams in *T* catchments. In a similar way, *T* has lower bifurcation ratio and Ro coefficient than nT catchments. According to Horton (1955), those low values are an indicator of the completeness of drainage of a basin, where low values mean a poorly dissected drainage basin, with a lower capacity of channel storage per unit of drainage area. Several authors have reported high bifurcation ratio for torrential catchments (Kanth and Hassan 2012; Rakesh et al. 2000); however, for the Colombian Andes, *T* catchments have lower mean values (3.50) compared to nT (4.78).

Considering the basin Geometry, all the analysed catchments are elongated, with values of elongation ratio < 0.47 , circularity ratio < 0.43 , and form factor < 0.18 . Since basins with elongated shapes have a hydrological response to rainfall events with flat peaks of long duration, it might be expected that nT catchments were more elongated than *T*. This is not the case of the results, since *T* catchments are relatively larger but with slightly more elongated shape (elongation ratio = 0.43 and form factor = 0.15) compared to nT catchments. This result might be explained by the lower bifurcation ratio values of *T*, since authors like Schumm (1956a, b) and Vinutha (2014) point out that basins with bifurcation ratio values < 1.0 have high relief and steep slope, which increase their peak flood response, compensating their elongated shapes.

In terms of drainage texture, *T* catchments show important differences with nT. It is well known that the hydrology of catchments changes significantly in response to changes in drainage texture (Horton 1932; Melton 1957; Schumm 1956a, b; Zavoianu 1985). *T* catchments show relatively lower values for drainage density, stream frequency and drainage intensity, indicating a poorly dissected terrain with a relatively slow hydrological response (Carlston 1963; Melton 1957). The infiltration number is inversely proportional to the infiltration capacity of soils in a basin. *T* catchments show very diverse values of infiltration number, with a lower mean value than nT, but also many outliers with high values. This

shows a bivariate distribution of the infiltration number in T catchments: those with high values are typical of low permeabilities of the soils, that contribute to an increase in the runoff and to higher flooding peaks. Considering the poor drainage development and low permeability of such T catchments, it is expected that their response to strong rainstorms corresponds to an accumulation of surface runoff that increases the sediment erosion power along the slopes and channels of the basin. T basins with low infiltration number, on the other hand, are constituted in soils with higher permeabilities, that may reduce the runoff in the short term, but when exposed to long-lasting rainfall events, are likely to fail and generate clusters of landslides, providing high volumes of sediments to the flow.

In a coherent way, T catchments show higher values of length of overland flow and constant of channel maintenance, highly correlated variables that are very important in the prediction capacity of the model. Length of overland flow is the rainfall run length on the ground surface before falling into definite channels (Horton 1955), and constant of channel maintenance is defined as the ratio between area and stream length, representing the area required to maintain one kilometre of drainage channel (Horton 1955). These higher values for T also indicate a higher availability of surface runoff, that may increase the sediment production along their area. In summary, according to the drainage texture results obtained, torrential susceptibility is characteristic of catchments with an elongated shape, low dissection, and varying infiltration capacity owing to higher runoff volumes of lower rate and high sediment production.

Additional to runoff and sediment production, relief characteristics of the basins play a fundamental role in torrential susceptibility. These parameters are crucial because they determine the overall steepness of the drainage basin, and consequently the hydraulic energy; but at the same time, they are an indicator of the intensity of the erosional process operating on its slopes (Schumm 1956a, b; Zavoianu 1985). All the relief characteristics for T catchment show higher values than nT , indicating high potential energy available in steep and long slopes, which increase the torrential susceptibility. Relief ratio, Melton index, and relief were identified as some of the best predictor parameters, indicating the gravitational energy of a catchment.

When comparing the four groups of morphometric parameters, there is a contrasting characteristic for both types of basins. According to basin geometry properties, T should have a slow hydrological response with flat peaks; however according to their drainage texture properties, they show a poor development of the drainage network that responds under intense rainstorms with high volumes, high sediment production, and a fast hydrologic response, associated with high levels of potential energy due to high relief characteristics. In conclusion, high relief, poorly developed drainage network, and high erosion capacity are the main factors that control the torrential susceptibility of catchments in the Colombian Andes, compensating their elongated shape.

The Melton index is a well-known and widely used slope index for distinguishing basins prone to torrential flows along the world (Bovis and Jakob 1999; Hungr et al. 2001; Jackson et al. 1987; Wilford et al. 2004). Thresholds suggested by Wilford et al. (2004) to differentiate between flood, debris floods and debris flow prone catchments in west central British Columbia, Canada are shown in Fig. 12, drawn in black lines, and the studied basins have been plotted in the diagram. According to this study, although Melton index is a good predictor for T catchments in Colombia, it is not useful when applied directly as a threshold, as in other regions has been proposed. Although T catchments show a higher mean value of Melton index than nT , their ranges are strongly overlapped.

Considering the feature selection procedures, only a handful of variables may be sufficient for prediction of torrential flow susceptibility. The ranking of relative variable

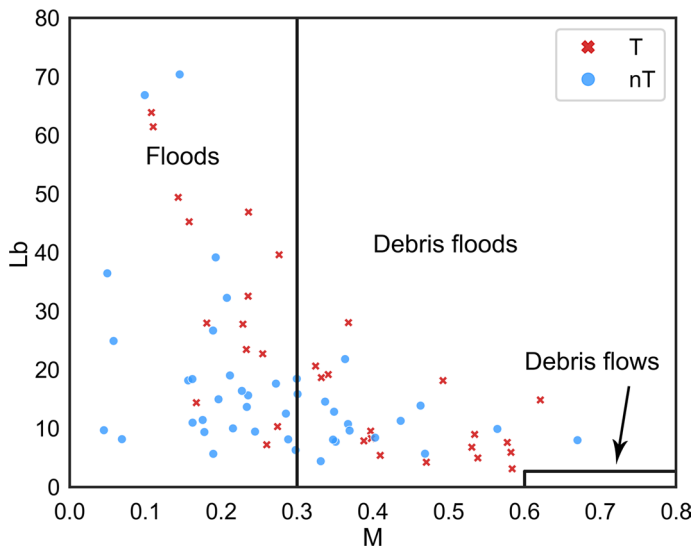


Fig. 12 M thresholds for basin susceptibility to flash floods, debris flows, and floods proposed by Wilford et al. (2004) with the T and nT catchments assessed in this study

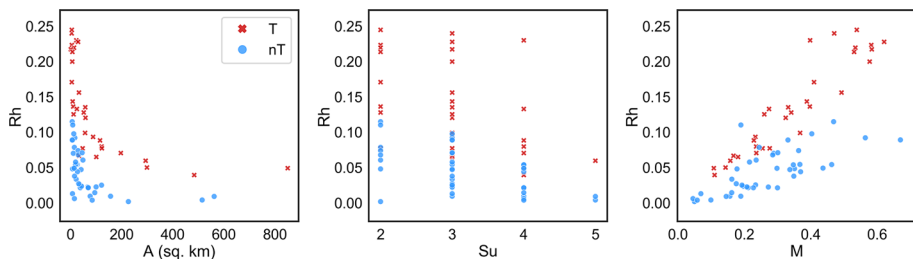


Fig. 13 Scatterplot of relief ratio against area, main stream order, and Melton index

importance was substantially different for all modelling techniques. However, there was consistency in the highest ranked predictors: relief ratio, constant of channel maintenance, relief, and form factor. According to this, T catchments are expected to have a high relief/length of catchment ratio, and a high constant of channel maintenance (and therefore low drainage density and high length of overland flow).

Regarding the area, several authors have reported a predominance of small areas for torrential catchments, such as Mizuyama (1982) who proposed a range from 0.1 to 100 km², and Rickenmann et al. (2008) who reported areas up to 25 km². Torrential catchment areas in the Colombian Andean range from very high (849.9 km²) to very small (1.33 km²), with a predominance of small areas, where 75% of T show values less than 100 km². But it is important to note that even in T basins with large areas, the values of relief ratio are higher compared to nT catchments (Fig. 13, left).

When visualising changes of main stream order and relief ratio (Fig. 13, center), it indicates a relief ratio threshold at which catchments change from an alluvial to torrential pattern as a function of their main stream order. Finally, although Melton index does not

permit a direct distinction between T and nT (Fig. 13, right), when visualised against relief ratio, it separates the groups in a clear way. T catchments have a steeper pattern, indicating higher values of relief ratio for similar values of Melton index. This confirms that relief ratio is a better indicator of torrentiality in the study area.

8 Conclusions

According to the results obtained in this study, the relief and drainage texture properties of catchments in the Colombian Andes are the most important when determining its torrential flow susceptibility. Within these groups of parameters, relief ratio and constant of channel maintenance are the highest ranked variables. Length of overland flow and Melton index show a good explaining capacity as well; however, they are highly correlated with constant of channel maintenance and relief ratio, respectively. Relief and form factor show moderate importance values to explain the torrential susceptibility, and they are not correlated to any other variable.

This analysis of morphometric parameters indicates that torrential basins in the tropical Andes are characterised by high relief, low infiltration capacity, and low dissection. In contrast to other studies in different environments, in the Colombian Andes they have a wide range of values for Area and Melton index, elongated shapes, and low values of drainage density and bifurcation ratio.

Although some morphometric parameters, such as area, perimeter, main stream length, main stream order and Melton index, are not identified in the selection feature analysis as good predictors, when analysed in a bivariate way with relief ratio, they provide a differentiation between T and nT catchments. All the T catchments have higher values of relief ratio than nT ones.

These results improve the general understanding of torrential occurrence in the Colombian Andes by identifying torrential prone catchments using morphometric analysis, a crucial step in the prioritisation of areas for detailed risk and hazard analysis.

Funding No funding was received for this project.

Availability of data and materials Dataset is provided.

Code availability Python code is provided.

Compliance with ethical standards

Conflict of interest All the authors manifest there are no conflict of interest.

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